

Treat the following questions:

I- Define data warehouse and explain its characteristics. (3pts)

II- Describe briefly the Extraction phase in ETL. (2pts)

III- The purpose of this exercise is to apply the Apriori algorithm on the following grocery store data, using as support and confidence thresholds $\text{minsup} = 33.3\%$ and $\text{minconf} = 60\%$.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

- Calculate the sets of k-itemsets candidates and frequent for each step of the algorithm.
- Show all generated association rules, by sorting these rules by decreasing confidence. (7pts)

IV- The general manager of an insurance and reinsurance company, located in many cities around the world, decides to analyze the incomes in its sales offices. He disposes some information about customers and policies in each of his offices, like: customer name, customer date of birth, customer sex, customer city, customer email, customer telephone, policy number, policy date of issue, policy type (live, house, car,...) policy monthly amount per \$, policy office, policy city, policy country. He desires to obtain information about the distribution of incomes per country, city, office, customer age, policy type and customer sex.

- Define the analysis axes and their hierarchies.
- Define the principal measures (aggregates).
- Draw star schema for datawarehouse.
- Draw the tree showing the multiple views of datawarehouse.
- Draw snowflake schema normalizing the dimension office.
- Add the necessary information and draw the constellation schema when the manager proposes to count and to calculate the total per \$, for later statistics, the distribution of unpaid policies per policy office, city office, and policy type.
- After creating DW, talk about 3 decisions that can be taken by the general manager. (8pts)